

Problem 5. (Optional) In class we introduced the derivative transformation $D : V_3 \rightarrow V_3$, defined by $D(f(t)) = f'(t)$. The derivative of a degree 2 polynomial has degree 1, so we can also view D as a linear transformation $V_2 \rightarrow V_1$.

- (a) Find the matrix representation of $D : V_2 \rightarrow V_1$ with respect to the bases $1, t, t^2$ for V_2 and $1, t$ for V_1 . Using this matrix and the one you found in Problem 4 (b), find the matrix representing the linear transformation $Q : V_2 \rightarrow V_2$ defined by $Q(f(t)) = (t - 1)f'(t)$.
- (b) Find the matrix representing the linear transformation $T : V_2 \rightarrow V_2$ defined by

$$T(f(t)) = (t - 1)f'(t) - 2f(t)$$

Find a basis for the kernel of this matrix, and use it to give a description of all polynomials $f(t)$ of degree at most 2 which satisfy the differential equation:

$$(t - 1)f'(t) = 2f(t)$$

- (c) We have alternative bases $1, t - 1$ for V_1 and $1, (t - 1), (t - 1)^2$ for V_2 . Find the matrix representations of M , D , Q , and T with respect to these bases. Does this make it easier to find the kernel of T ?

Solution. (a) We have $D(1) = 0$, $D(t) = 1$, $D(t^2) = 2t$. In the coordinate system we have on V_1 , these correspond to vectors

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

and so our matrix for D is

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

The matrix for $M : V_1 \rightarrow V_2$ is

$$\begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix}$$

The transformation Q is the composition $M \circ D$, which corresponds to the matrix product

$$\begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 2 \end{bmatrix}$$

- (b) T sends f to $Q(f) - 2 \cdot f$. The matrix corresponding to this is the matrix for Q minus $2I_3$, or

$$\begin{bmatrix} 0 & -1 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 2 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -2 & -1 & 0 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

To find the kernel of this matrix, we apply row operations to bring it to reduced row echelon form:

$$\begin{bmatrix} -2 & -1 & 0 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} -2 & -1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} -2 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Assigning variables x, y, z to the columns, z is the free variable, and we have $x = z$, $y = -2z$. Thus a general element of the kernel has the form

$$\begin{bmatrix} z \\ -2z \\ z \end{bmatrix} = z \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

and so the kernel is

$$\text{span} \left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \right\}$$

Elements of the kernel of this matrix correspond under our chosen basis for V_2 to elements $f \in V_2$ with $T(f) = 0$, or in other words, polynomials $f \in V_2$ satisfying

$$(t-1)f'(t) = 2f(t)$$

All such polynomials are scalar multiples of

$$t^2 - 2t + 1 = (t-1)^2$$

- (c) As a side remark here, $1, (t-1), (t-1)^2$ is where the transformation S from the previous problem takes $1, t, t^2$. Invertible linear transformations always take one basis to another, so this is a way of seeing that these polynomials form a basis. The same thing works for $1, t-1$ as a basis of V_1 .

To express $M : V_1 \rightarrow V_2$ in terms of these bases, we see where M sends each basis element in V_1 and express it as a linear combination of the V_2 basis elements. In this case the answer is simple:

$$M(1) = t - 1 \rightsquigarrow \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$M(t-1) = (t-1)^2 \rightsquigarrow \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Thus the matrix for M is

$$\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

To find the matrix for D , we see what happens when we take the derivative of the V_2 basis elements:

$$D(1) = 0 \rightsquigarrow \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$D(t-1) = 1 \rightsquigarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$D((t-1)^2) = 2(t-1) \rightsquigarrow \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

and so our matrix is

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Again the matrix for Q is the product

$$\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

We again find the matrix for T by subtracting $2I_3$:

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

For this matrix, it is fairly straightforward to see that the kernel is

$$\text{span} \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Translating back into elements of V_2 using our new basis $1, (t-1), (t-1)^2$, we conclude as before that the solutions to the differential equation in V_2 are exactly scalar multiples of $(t-1)^2$.

Remark. For any $n \in \mathbb{N}$ and any $a \in \mathbb{R}$, the solutions to the differential equation

$$(t-a)f'(t) = n \cdot f(t)$$

are exactly the scalar multiples of the polynomial $(t-a)^n$.